

Solving n equations
 n variables

Expect
Single
Unique
Solution

$$Ax = b$$

Theoretically

$$x = \frac{b}{A} = A^{-1}b$$

↖ on left

Fact: $A_{n \times n}$ matrix

A^{-1} exists iff $\det A \neq 0$

For $n=2$ $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\det A = ad - bc$

A^{-1} exists iff $\text{rank } A = n$

System of Equations

∞ Solutions

$$Ax = b$$

General Solution

$$x_0 + v$$

where $v \in \text{Null}(A) = \{v \mid Av = 0\}$

Reason: if $A \cdot x_0 = b$

and $A \cdot v = 0$

$$\begin{aligned} \text{then } A(x_0 + v) &= Ax_0 + Av \\ &= b + 0 = b \end{aligned}$$

Got x_0 from least squares

- Check $Ax_0 = b$

v is a linear combination
of null vectors

Spanned last rows of V^t in SVD

V_h in Python

Matrix Rank

"dimension of matrix"

Null Free variables in NullSpace

= # rows in SVD

$$A v_i = 0 \quad \text{with } S_i = 0$$

$$\text{rank} + \dim \text{Null} = \# \text{columns}$$

rank $\sim$ # of actual constraints

rows = apparent # constraints

SVD = Singular Value Decomposition

$$A = U S V^t$$

u_i columns of U

v_i columns of V

$$A v_i = S_i u_i$$

S_i diagonal of S

$\uparrow$

Matrix

$\uparrow$

Number

$S_i = \text{Singular Value}$

When $S_i = 0$ then v_i in NullSpace